Web App QA & Debug Report

Candidate: Gowri **App tested:** Conduit (RealWorld Angular demo), https://demo.realworld.show **Date:** 9 November 2026 **Browser:** Google Chrome on Windows

1. Summary

I tested the main user flows (sign-up, login, create, edit and delete articles, comments, refresh, log out) and ran PageSpeed Insights / Lighthouse on the site. Core features work, but I found **5 issues**: two in form behaviour, and three quality problems (layout shift, slow loading, accessibility) that would matter in production. Note: the demo link in the assignment brief (demo.realworld.io) is no longer available (it returns "404 NoSuchBucket"), and older demo versions had no working backend, so I used the maintained demo at demo.realworld.show.

2. Bug table

#	Title / Summary	Steps to Reproduce	Expected vs Actual	Severity	Suspected Cause
1	**No feedback on empty sign-up form**	1. Open Sign up. 2. Leave username, email and password empty. 3. Try to click the Sign up button.	**Expected:** messages such as "Username is required". **Actual:** the button is greyed out and no message explains why.	Low	The frontend disables the submit button but does not show field-level validation messages.
2	**Duplicate sign-up shows no error**	1. Sign up with a new username and email. 2. Log out. 3. Sign up again with the same username and email (a different password also tried).	**Expected:** an error such as "username/email has already been taken". **Actual:** no error message appeared on screen. (Server status code not recorded in this test.)	Medium (High if the server accepts duplicates)	Either the server does not enforce unique usernames/emails, or the frontend does not display the server's error. Needs a check of the network response to tell which.
3	**Layout shifts while the page loads**	1. Open the home page. 2. Run PageSpeed Insights on it. 3. Look at Cumulative Layout Shift (CLS) and the tag area of the first article in a narrow window.	**Expected:** CLS of 0.1 or less. **Actual:** CLS 0.29 (real-user data) and 0.364 (second test run); Core Web Vitals assessment: Failed. In a narrow window the "python" tag wrapped and overlapped the divider line.	Medium	Late-loading web fonts and content that appears without reserved space (PageSpeed flags "Font display" and "Layout shift culprits").
4	**Slow initial load**	1. Run PageSpeed Insights on the home page. 2. Check the load metrics and Insights list.	**Expected:** LCP 2.5 s or less and a fast server response. **Actual:** LCP 2.7 s, FCP 2.1 s, TTFB 1.2 s; render-blocking requests (est. 260 ms to 800 ms saving); 31 KiB unused JavaScript; weak cache lifetimes.	Medium	Render-blocking CSS/JS, unused JavaScript, short cache lifetimes and a slow server response.
5	**Poor accessibility (score 67/100)**	1. Run PageSpeed Insights / Lighthouse. 2. Open the Accessibility section and read the failed audits.	**Expected:** score of 90 or more. **Actual:** 67. Failures: images without alt text; links without a discernible name; insufficient colour contrast; no lang attribute on html; no main landmark; accessibility tree not well-formed.	Medium	Markup was built without accessibility attributes (alt text, link labels, lang, landmarks) and with low-contrast colours.

Severity scale: Critical, High, Medium, Low. Performance figures: real-user data is the 28-day aggregate for the whole site; lab figures come from my own PageSpeed runs and vary from run to run.

3. Root-cause analysis: layout shift (Issue 3)

What is happening. While the home page loads, content moves around on screen. PageSpeed measures this as a Cumulative Layout Shift of 0.29 for real users and 0.364 in my test run, where 0.1 or less is considered good, so the site fails the Core Web Vitals assessment. In a narrow window I also saw the "python" tag of the first article wrap onto a second line and overlap the divider below it. **Why it happens.** Layout shift happens when something is drawn first and then changes size or position. PageSpeed's Insights list shows a "Font display" warning, which means text is first shown in a fallback font and then re-drawn when the web font arrives, changing the size of lines and tags. The article list also appears after data is fetched from the API, and if space is not reserved for it, everything below it is pushed down. Avatar images without fixed dimensions can cause the same effect. **How I would fix it.** I would load fonts with a matching fallback font and a size-adjust setting (or preload the font), give images and cards fixed width and height or a minimum height, show a placeholder skeleton while articles load, and allow tags to wrap inside a container that has enough height. I would then re-run PageSpeed to confirm CLS is 0.1 or lower. *Note: I identified these as the most likely causes from the PageSpeed Insights list; confirming the exact elements requires opening the "Layout shift culprits" item or recording a trace in Chrome DevTools.*

4. Things I checked that worked correctly

• Invalid email at sign-up shows an "invalid email" message.
• Wrong password at login shows a generic "invalid credentials" message. This is acceptable: it does not reveal whether the username exists.
• Creating an article with empty fields shows an error message.
• Editing and deleting an article, and adding and deleting a comment, all work.
• The user stays logged in after refreshing the page.
• Keyboard focus is visible when pressing Tab on the home page.
• PageSpeed Best Practices score is 100.

5. Evidence (screenshots)

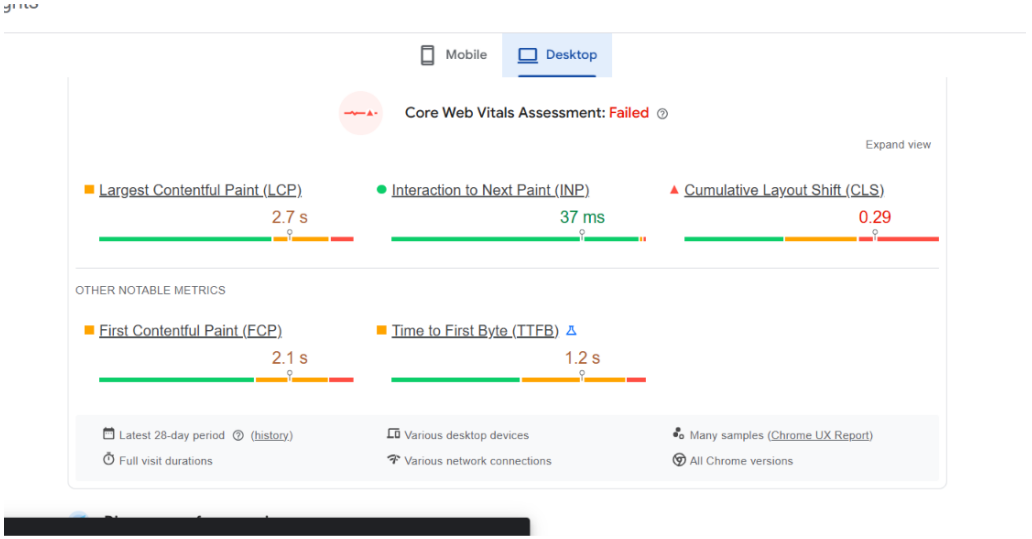


Figure 1 - PageSpeed Insights: Core Web Vitals failed (LCP 2.7 s, CLS 0.29, TTFB 1.2 s). Supports Issues 3 and 4.

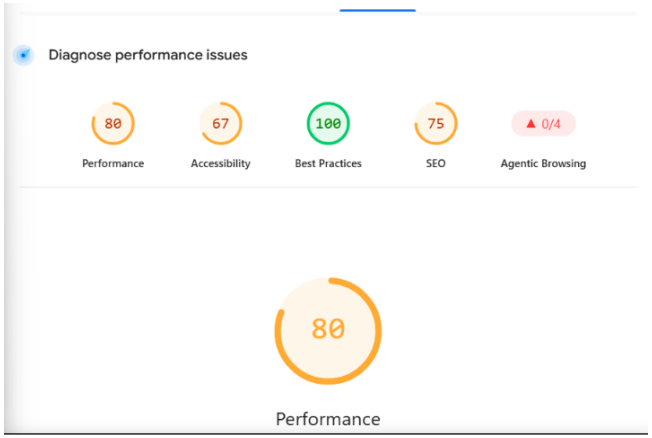


Figure 2 - Lighthouse scores: Performance 80, Accessibility 67, Best Practices 100, SEO 75. Supports Issue 5.

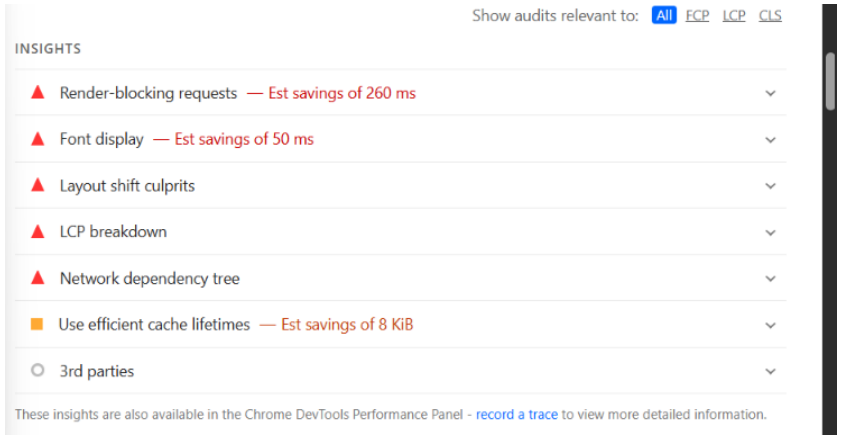


Figure 3 - PageSpeed Insights list: render-blocking requests, font display, layout shift culprits. Supports Issues 3 and 4.

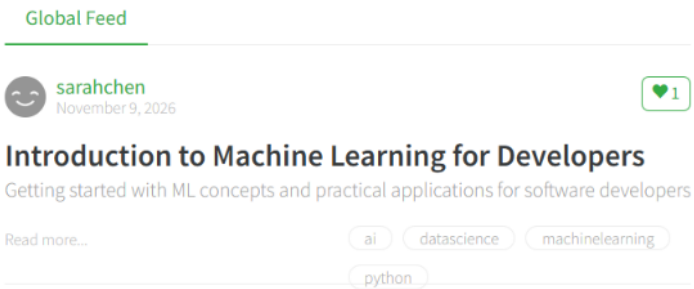


Figure 4 - The "python" tag wrapping and overlapping the divider line in a narrow window. Supports Issue 3.